

Introduction to Claude Code

Weekend 1, Saturday parallel session

CAS Building ML/AI Applications, ETH Zurich

After the weekend 1 lecture of From Data to Decisions, by Francesco Caperna and Deli Lin

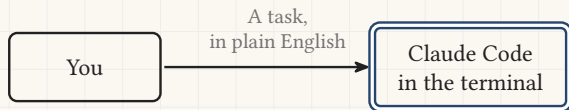
Saturday 5 September 2026

1

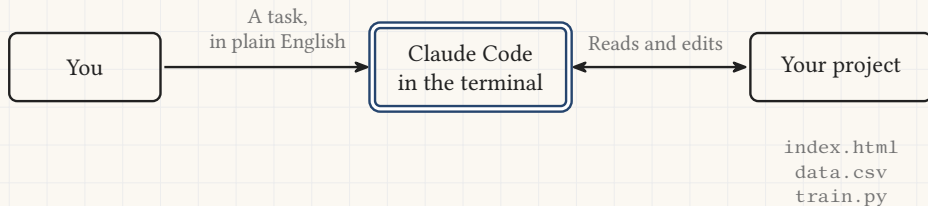
Section 1

Claude Code

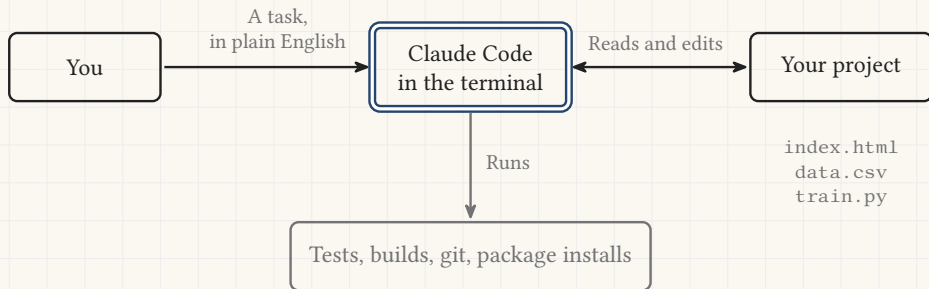
What Claude Code is



What Claude Code is

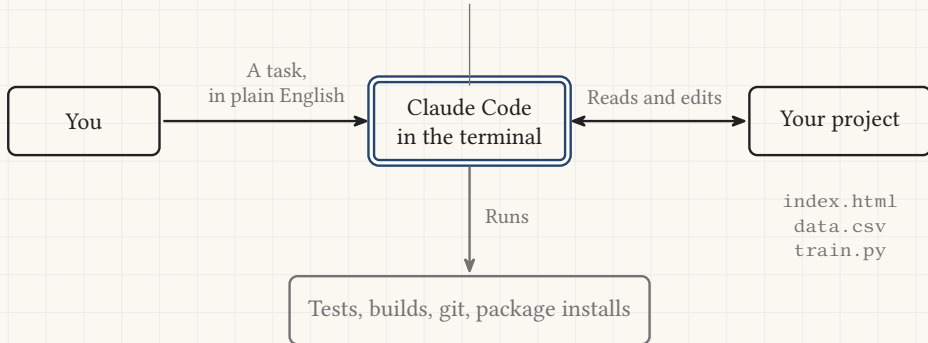


What Claude Code is



What Claude Code is

It also runs in VS Code, in JetBrains and in a desktop app. We stay in the terminal, where every change is visible before it lands.



Why this course teaches it

It does not solve every problem. It lets you try more solutions in the same hour.

Why this course teaches it

It does not solve every problem. It lets you try more solutions in the same hour.

The coding exercises
of every weekend

A stub to fill in
reads faster with a
reader beside you

Why this course teaches it

It does not solve every problem. It lets you try more solutions in the same hour.

The coding exercises
of every weekend

A stub to fill in
reads faster with a
reader beside you

The weekend project,
built end to end

A full pipeline is
more code than a
weekend has hours

Why this course teaches it

It does not solve every problem. It lets you try more solutions in the same hour.

The coding exercises
of every weekend

A stub to fill in
reads faster with a
reader beside you

The weekend project,
built end to end

A full pipeline is
more code than a
weekend has hours

Your own work,
after the course

The licence outlives the
weekend it was bought for

What you need on your laptop

Three things. Two of them you install once, and one is already there.

VS Code

A free text editor
from Microsoft, and
the window you
look at the files in

What you need on your laptop

Three things. Two of them you install once, and one is already there.

VS Code

A free text editor
from Microsoft, and
the window you
look at the files in

A terminal

A window where you
type commands. It is
already on your machine

What you need on your laptop

Three things. Two of them you install once, and one is already there.

VS Code

A free text editor from Microsoft, and the window you look at the files in

A terminal

A window where you type commands. It is already on your machine

Claude Code

The program itself. It runs inside the terminal

What you need on your laptop

Three things. Two of them you install once, and one is already there.

VS Code

A free text editor from Microsoft, and the window you look at the files in

A terminal

A window where you type commands. It is already on your machine

Claude Code

The program itself. It runs inside the terminal

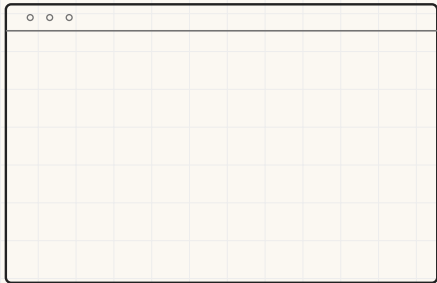
Every step is written out in the setup guide, with a button that copies each command, so nothing has to be typed by hand.

eth-bmai-hs26.github.io/BMAI-PAGE/guides/claude-code-setup.html



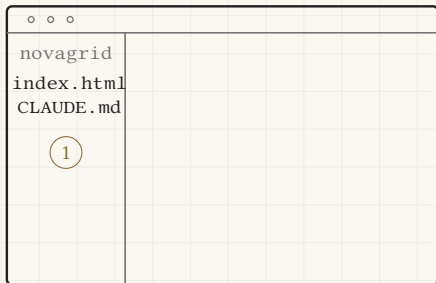
VS Code

A free text editor made by Microsoft. Three parts matter this morning.



VS Code

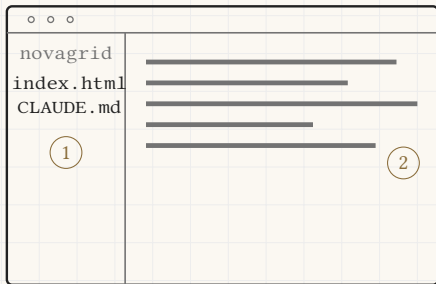
A free text editor made by Microsoft. Three parts matter this morning.



- The strip on the left lists the files of your project.

VS Code

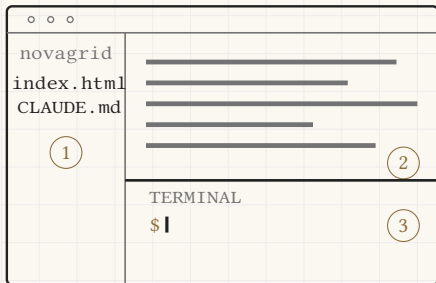
A free text editor made by Microsoft. Three parts matter this morning.



- The strip on the left lists the files of your project.
- The middle shows the one you have open.

VS Code

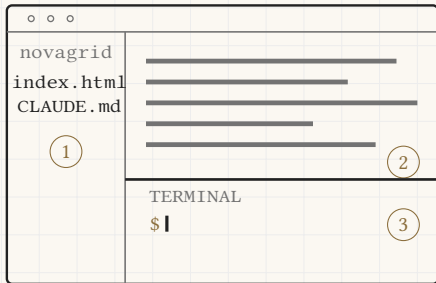
A free text editor made by Microsoft. Three parts matter this morning.



- The strip on the left lists the files of your project.
- The middle shows the one you have open.
- The panel along the bottom is a terminal, and that is where Claude Code runs.

VS Code

A free text editor made by Microsoft. Three parts matter this morning.



- The strip on the left lists the files of your project.
- The middle shows the one you have open.
- The panel along the bottom is a terminal, and that is where Claude Code runs.

It is a free download:

code.visualstudio.com/download

Step 1 of the setup guide walks through it, click by click, for a Mac and for Windows.

Opening a terminal

On a Mac

- Press Cmd + Space.
- Type Terminal, press Enter.

A terminal window icon consisting of three small circles (two red, one green) arranged in a horizontal line.

you@macbook % |

On Windows

Opening a terminal

On a Mac

- Press Cmd + Space.
- Type Terminal, press Enter.

ooo

you@macbook % |

On Windows

- Press Win + X.
- Choose **Windows PowerShell**, or **Terminal**.

ooo

PS C:\Users\You> |

Opening a terminal

On a Mac

- Press Cmd + Space.
- Type Terminal, press Enter.

○ ○ ○

you@macbook % |

On Windows

- Press Win + X.
- Choose **Windows PowerShell**, or **Terminal**.

○ ○ ○

PS C:\Users\You> |

- You type one line and press Enter. Nothing happens until you do.
- You cannot click on anything in there. Use the arrow keys.
- Esc interrupts Claude. exit closes it.

Installing it, and signing in

ooo

macOS, Linux, WSL

```
$ curl -fsSL https://claude.ai/install.sh | bash
```

Windows PowerShell

```
$ irm https://claude.ai/install.ps1 | iex
```

```
$ claude --version
```

```
2.1.211 (Claude Code)
```

Installing it, and signing in

○ ○ ○

macOS, Linux, WSL

```
$ curl -fsSL https://claude.ai/install.sh | bash
```

Windows PowerShell

```
$ irm https://claude.ai/install.ps1 | iex
```

```
$ claude --version  
2.1.211 (Claude Code)
```

- **Do not type these.** Open the setup guide and press the copy button beside each one.

Installing it, and signing in

○ ○ ○

macOS, Linux, WSL

```
$ curl -fsSL https://claude.ai/install.sh | bash
```

Windows PowerShell

```
$ irm https://claude.ai/install.ps1 | iex
```

```
$ claude --version
```

```
2.1.211 (Claude Code)
```

- **Do not type these.** Open the setup guide and press the copy button beside each one.
- You were sent an **email inviting you to join the course's Claude team**. Accept it, then sign in with that account the first time you run `claude`.

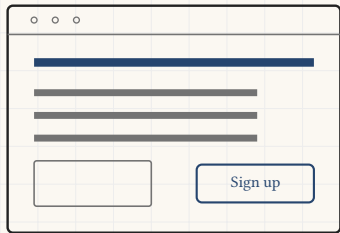
2

Section 2

Git, the safety net

Why Git

You ask for a small change. Claude edits your files, and the page comes back different.



Before: everything works

Why Git

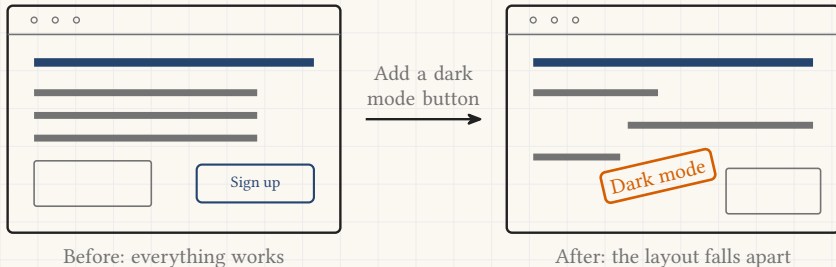
You ask for a small change. Claude edits your files, and the page comes back different.



Before: everything works

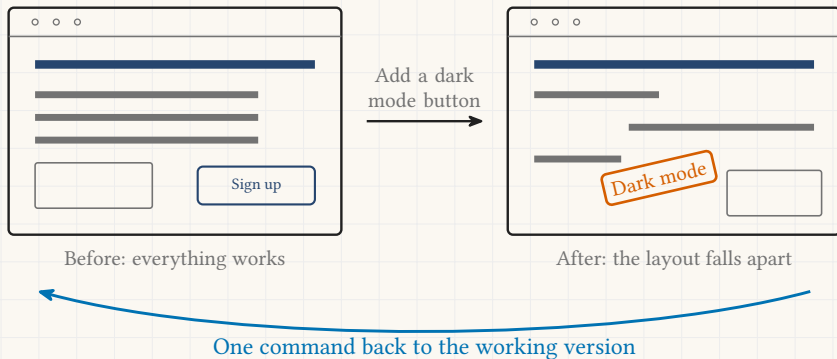
Why Git

You ask for a small change. Claude edits your files, and the page comes back different.



Why Git

You ask for a small change. Claude edits your files, and the page comes back different.



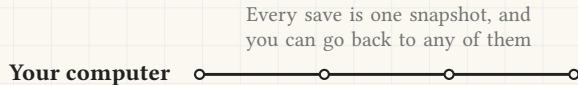
How Git works

Git keeps a snapshot of the project every time you save one, and nothing is ever written over.



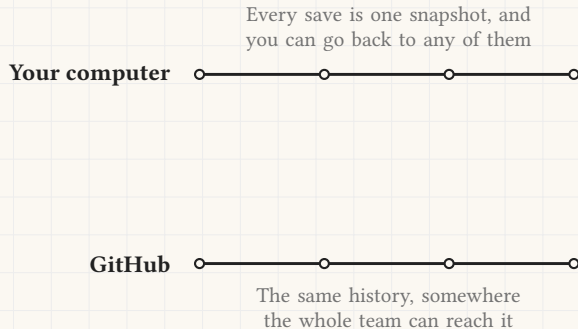
How Git works

Git keeps a snapshot of the project every time you save one, and nothing is ever written over.



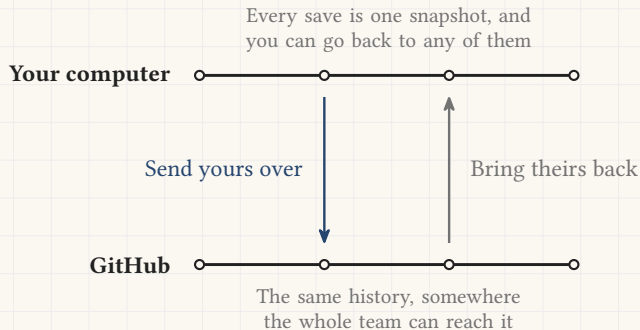
How Git works

Git keeps a snapshot of the project every time you save one, and nothing is ever written over.



How Git works

Git keeps a snapshot of the project every time you save one, and nothing is ever written over.



You never type any of this

Four words name the four moves. You say them in English, Claude runs them, and it reports every command it ran.

- `add`, choose what goes into the snapshot
- `commit`, take the snapshot
- `push`, send it to GitHub
- `pull`, bring back what others sent

You never type any of this

Four words name the four moves. You say them in English, Claude runs them, and it reports every command it ran.

- add, choose what goes into the snapshot
- commit, take the snapshot
- push, send it to GitHub
- pull, bring back what others sent

o o o

> Save what we have and put it on GitHub.

Running git add .

Running git commit -m "feat: add hero"

Running git push

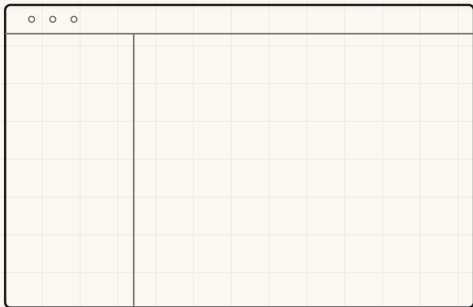
Done. 1 file changed.

3

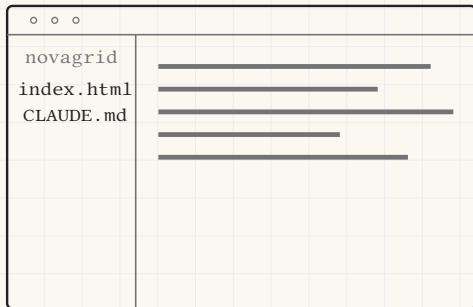
Section 3

A session from scratch

Step 1: the project folder and its terminal

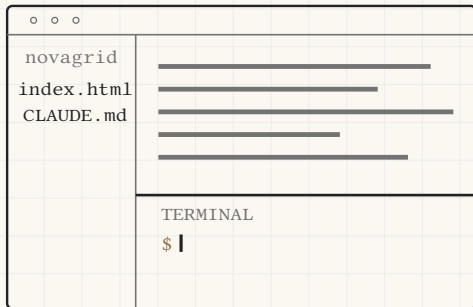


Step 1: the project folder and its terminal



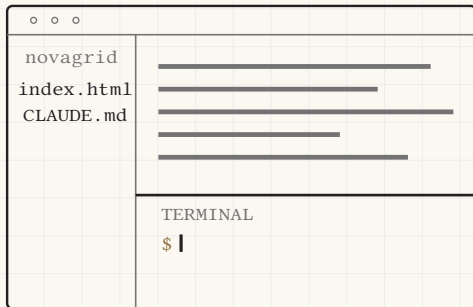
- Open the folder in VS Code: **File**, then **Open Folder**.

Step 1: the project folder and its terminal



- Open the folder in VS Code: **File**, then **Open Folder**.
- Open the built in terminal: **Terminal**, then **New Terminal**.
- Or press `Ctrl + ``.

Step 1: the project folder and its terminal



- Open the folder in VS Code: **File**, then **Open Folder**.
- Open the built in terminal: **Terminal**, then **New Terminal**.
- Or press `Ctrl + ``.
- Every command in this lecture is typed in that panel.

Step 2: start Claude Code

o o o

\$ claude

Welcome to Claude Code!

>

Step 2: start Claude Code

o o o

\$ claude

Welcome to Claude Code!

>

- The prompt is now Claude's. Type in plain English.

Step 2: start Claude Code

o o o

\$ claude

Welcome to Claude Code!

>

- The prompt is now Claude's. Type in plain English.
- It starts in the folder you are standing in, so it sees that project and no other.

Step 2: start Claude Code

o o o

\$ claude

Welcome to Claude Code!

>

- The prompt is now Claude's. Type in plain English.
- It starts in the folder you are standing in, so it sees that project and no other.
- This is the one command in this lecture you type yourself. The setup guide has a copy button for it.

Step 3: ask it to set the project up

ooo

> Set this folder up so my work is tracked, and save a first snapshot.

Running `git init`

Running `git add .`

Running `git commit -m "chore: initial commit"`

Done. You can come back to this point at any time.

Step 3: ask it to set the project up

o o o

> Set this folder up so my work is tracked, and save a first snapshot.

Running git init

Running git add .

Running git commit -m "chore: initial commit"

Done. You can come back to this point at any time.

- You did not type a git command. You said what you wanted, and it told you what it ran.

Step 3: ask it to set the project up

o o o

> Set this folder up so my work is tracked, and save a first snapshot.

Running git init

Running git add .

Running git commit -m "chore: initial commit"

Done. You can come back to this point at any time.

- You did not type a git command. You said what you wanted, and it told you what it ran.
- From here on the whole of git is two sentences: save a snapshot, and put it on GitHub.

The command menu

Type a forward slash to open the command menu. Keep typing to filter it, then press enter.

ooo

> /

/help	show all commands
/clear	clear the conversation
/model	choose the model
/context	show what is filling the window
...	

The command menu

Type a forward slash to open the command menu. Keep typing to filter it, then press enter.

ooo

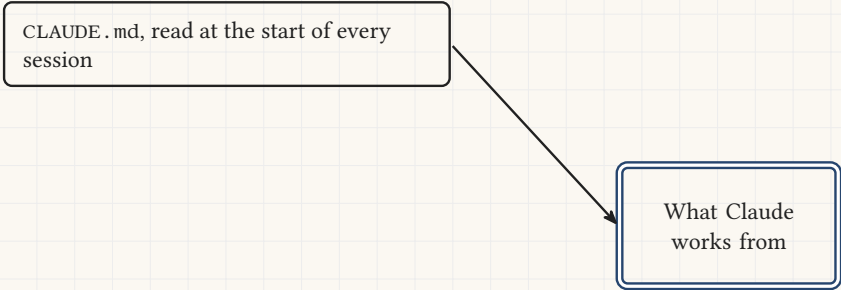
> /

/help	show all commands
/clear	clear the conversation
/model	choose the model
/context	show what is filling the window
...	

Everything Claude Code can be told to do about itself lives behind that slash.

Five ways to give Claude context

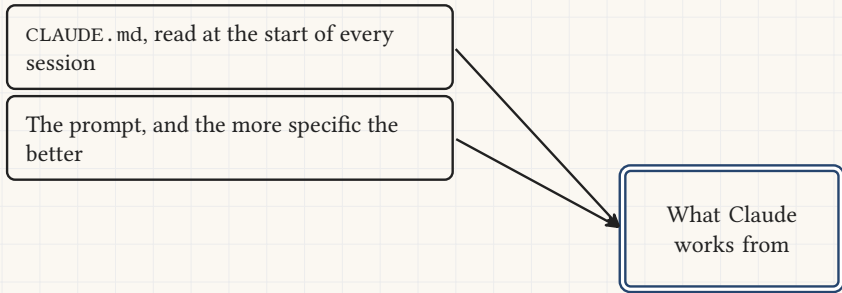
CLAUDE.md, read at the start of every session



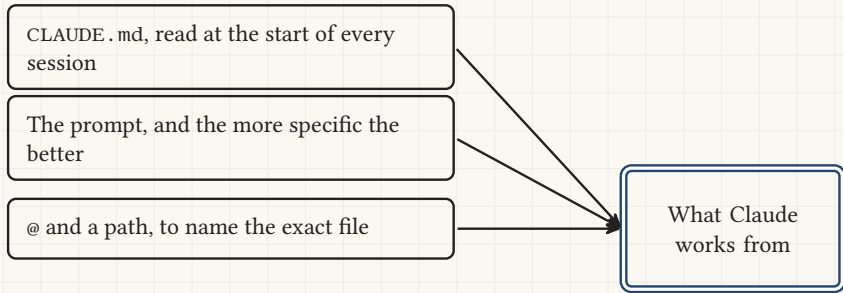
```
graph TD; A[CLAUDE.md, read at the start of every session] --> B[What Claude works from];
```

What Claude works from

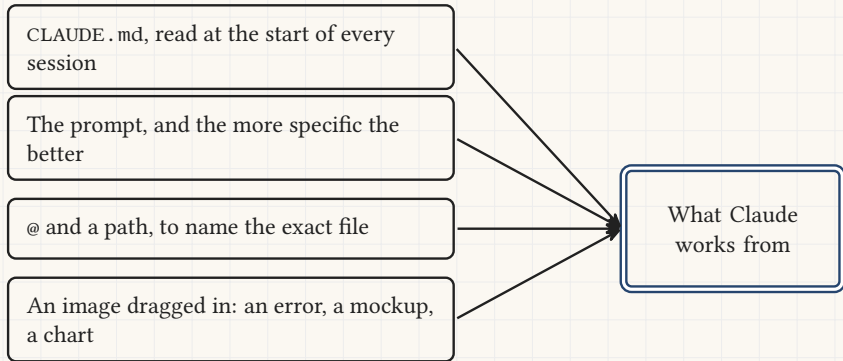
Five ways to give Claude context



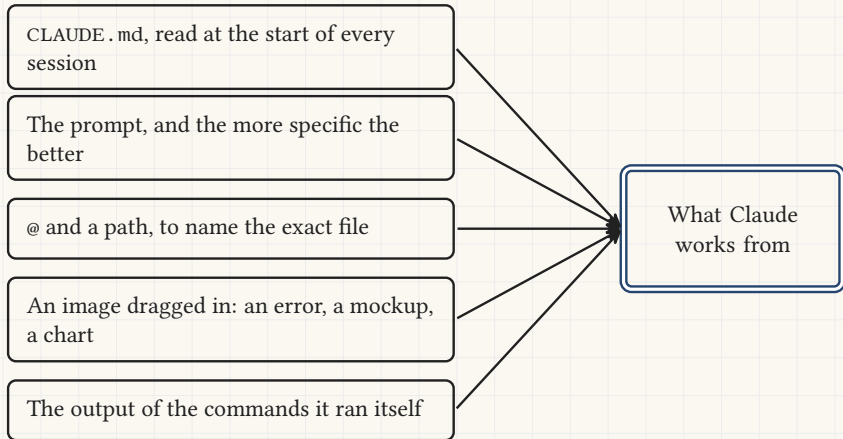
Five ways to give Claude context



Five ways to give Claude context



Five ways to give Claude context



CLAUDE.md, the file it reads first

CLAUDE.md

Commands

- Run tests: `pytest`
- Start app: `python main.py`

Project structure

- `src/` core code
- `data/` raw datasets, do not edit

Conventions

- Use type hints everywhere

CLAUDE.md, the file it reads first

CLAUDE.md

Commands

- Run tests: `pytest`
- Start app: `python main.py`

Project structure

- `src/` core code
- `data/` raw datasets, do not edit

Conventions

- Use type hints everywhere

- How to build, test and run the project.
- Where things are, and which files matter.
- The conventions, and the things to avoid.

CLAUDE.md, the file it reads first

CLAUDE.md

Commands

- Run tests: `pytest`
- Start app: `python main.py`

Project structure

- `src/` core code
- `data/` raw datasets, do not edit

Conventions

- Use type hints everywhere

- How to build, test and run the project.
- Where things are, and which files matter.
- The conventions, and the things to avoid.

o o o

> /init

Analyzing your project...

Created CLAUDE.md

CLAUDE.md, the file it reads first

CLAUDE.md

Commands

- Run tests: `pytest`
- Start app: `python main.py`

Project structure

- `src/` core code
- `data/` raw datasets, do not edit

Conventions

- Use type hints everywhere

- How to build, test and run the project.
- Where things are, and which files matter.
- The conventions, and the things to avoid.

ooo

> /init

Analyzing your project...

Created CLAUDE.md

It writes a first version. Correct it from there.

Pointing at a file, and making it remember

o o o

> Explain what @src/main.py does

A path after @ pulls that file in.

Pointing at a file, and making it remember

o o o

> Explain what @src/main.py does

A path after @ pulls that file in.

o o o

> Remember that we always use pytest
Saved to memory.

Ask it to remember, and the note survives the session.

Pointing at a file, and making it remember

o o o

> Explain what @src/main.py does

A path after @ pulls that file in.

o o o

> Remember that we always use pytest
Saved to memory.

Ask it to remember, and the note survives the session.

o o o

> /memory

Lists your memory files.
Select one to open it.

What it remembers is a plain text file you can edit.

4

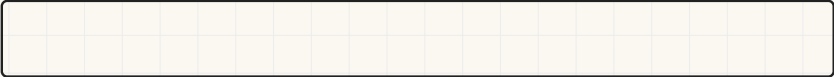
Section 4

Getting more out of it

The context window

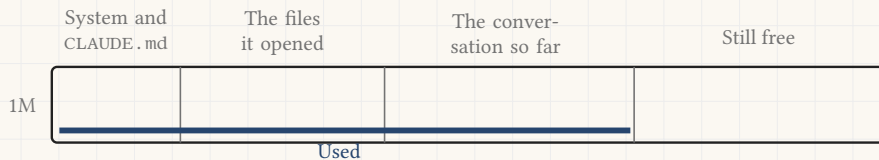
Your code, the files it opened and the whole conversation share one window, measured in tokens.

1M



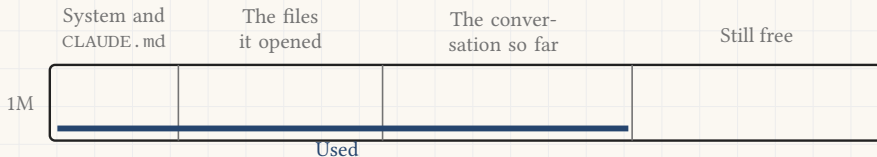
The context window

Your code, the files it opened and the whole conversation share one window, measured in tokens.



The context window

Your code, the files it opened and the whole conversation share one window, measured in tokens.



ooo

> /context

Context: 700k / 1M tokens used (70%)

Clearing and compacting

o o o

> /clear

Context cleared. Starting fresh.

	Empty
--	-------

Kept

Everything goes except CLAUDE.md, which is read again. Use it when the next task is unrelated.

Clearing and compacting

ooo

```
> /clear
```

Context cleared. Starting fresh.

	Empty
--	-------

Kept

Everything goes except CLAUDE.md, which is read again. Use it when the next task is unrelated.

ooo

```
> /compact
```

Summarizing conversation...

Context compacted, tokens freed.

		Freed
--	--	-------

Kept Summary

The conversation becomes a summary of itself. Use it when the next task continues this one.

Choosing the model

Fable, for work larger than one sitting

Choosing the model

Fable, for work larger than one sitting

Opus, for the complex and the tricky

Choosing the model

Fable, for work larger than one sitting

Opus, for the complex and the tricky

Sonnet, fast and balanced

Choosing the model

Fable, for work larger than one sitting

Opus, for the complex and the tricky

Sonnet, fast and balanced

Haiku, quickest and cheapest

Faster, and cheaper



Choosing the model

Fable, for work larger than one sitting

Opus, for the complex and the tricky

Sonnet, fast and balanced

Haiku, quickest and cheapest

Faster, and cheaper
↓

o o o

```
> /model
```

```
Fable      most capable, chosen
```

```
Opus       complex and tricky
```

```
Sonnet     fast and balanced
```

```
Haiku      quickest
```

Choosing the model

Fable, for work larger than one sitting

Opus, for the complex and the tricky

Sonnet, fast and balanced

Haiku, quickest and cheapest

Faster, and cheaper
↓

ooo

```
> /model
Fable    most capable, chosen
Opus     complex and tricky
Sonnet   fast and balanced
Haiku    quickest
```

Match the model to the task. A footer does not need the most capable one. A subtle bug does.

Three ways to work

Edit

It reads and edits your files to carry the task out. Best when the task is clear.

The default

Three ways to work

Edit

It reads and edits your files to carry the task out. Best when the task is clear.

The default

Plan

It researches and proposes the steps, and changes nothing until you approve.

`/plan`, or Shift + Tab

Three ways to work

Edit

It reads and edits your files to carry the task out. Best when the task is clear.

The default

Plan

It researches and proposes the steps, and changes nothing until you approve.

`/plan`, or Shift + Tab

Effort

It reasons longer before acting. For a tricky bug or a design decision.

`/effort`

Three ways to work

Edit

It reads and edits your files to carry the task out. Best when the task is clear.

The default

Plan

It researches and proposes the steps, and changes nothing until you approve.

`/plan`, or `Shift + Tab`

Effort

It reasons longer before acting. For a tricky bug or a design decision.

`/effort`

Shift + Tab cycles how much Claude may do without asking: every edit confirmed, edits accepted automatically, or plan mode.

Writing your own command

A Markdown file in `.claude/commands/` becomes a command you can call by its own name.

`.claude/commands/qa-check.md`

Check `index.html` for leftover placeholder text, dead links and missing alt attributes. Report each one with a severity.

Writing your own command

A Markdown file in `.claude/commands/` becomes a command you can call by its own name.

`.claude/commands/qa-check.md`

Check `index.html` for leftover placeholder text, dead links and missing alt attributes. Report each one with a severity.



ooo

```
> /qa-check
Reading index.html...
3 issues found.
```

Writing your own command

A Markdown file in `.claude/commands/` becomes a command you can call by its own name.

```
.claude/commands/qa-check.md
```

```
Check index.html for leftover  
placeholder text, dead links and  
missing alt attributes. Report  
each one with a severity.
```



```
ooo
```

```
> /qa-check  
Reading index.html...  
3 issues found.
```

Good for a check you run every time, written down once so nobody has to remember it.

Commands worth knowing

Knowing that the features exist is what matters. Ask in plain English for the rest.

- `/help`, list every command
- `/init`, write a first `CLAUDE.md`
- `/memory`, read and edit what it remembers
- `/context`, show what fills the window
- `/clear`, start the conversation again
- `/compact`, summarize it to free space
- `/model` and `/effort`, pick the engine
- `/review`, review a diff or a pull request

Commands worth knowing

Knowing that the features exist is what matters. Ask in plain English for the rest.

- `/help`, list every command
- `/init`, write a first `CLAUDE.md`
- `/memory`, read and edit what it remembers
- `/context`, show what fills the window
- `/clear`, start the conversation again
- `/compact`, summarize it to free space
- `/model` and `/effort`, pick the engine
- `/review`, review a diff or a pull request

ooo

> How do I free up the context?

Use `/compact` to summarize the conversation.

Two things to know exist*

Subagents

- For a big task, Claude can hand parts of the work to helpers that run on their own.
- Each one carries its own context, so the main conversation stays small.
- Manage them with `/agents`.

MCP

Two things to know exist*

Subagents

- For a big task, Claude can hand parts of the work to helpers that run on their own.
- Each one carries its own context, so the main conversation stays small.
- Manage them with `/agents`.

MCP

- The Model Context Protocol connects Claude Code to tools outside your folder.
- Databases, GitHub, Slack, a browser, and so on.
- Add a server with `claude mcp add`, inspect them with `/mcp`.

Two things to know exist*

Subagents

- For a big task, Claude can hand parts of the work to helpers that run on their own.
- Each one carries its own context, so the main conversation stays small.
- Manage them with `/agents`.

Neither is needed for this morning's exercise.

MCP

- The Model Context Protocol connects Claude Code to tools outside your folder.
- Databases, GitHub, Slack, a browser, and so on.
- Add a server with `claude mcp add`, inspect them with `/mcp`.

The exercise: Operation Midnight Launch

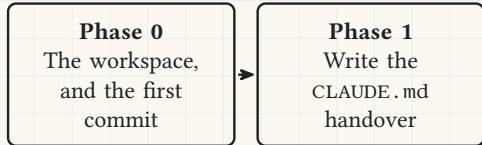
It is 11 at night. The freelancer who was building the investor demo page has vanished, the investors arrive at nine, and you have one engineer left.

Phase 0

The workspace,
and the first
commit

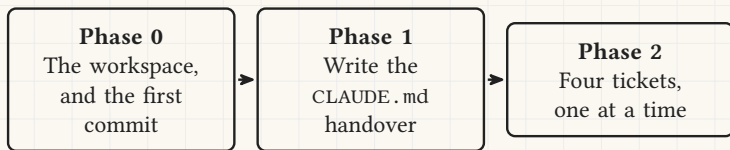
The exercise: Operation Midnight Launch

It is 11 at night. The freelancer who was building the investor demo page has vanished, the investors arrive at nine, and you have one engineer left.



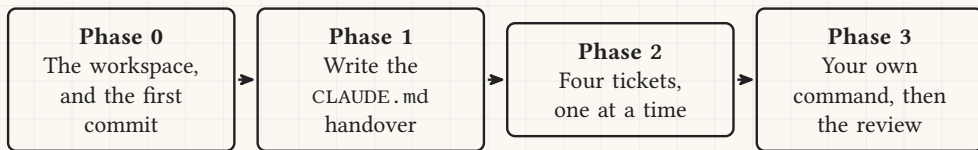
The exercise: Operation Midnight Launch

It is 11 at night. The freelancer who was building the investor demo page has vanished, the investors arrive at nine, and you have one engineer left.



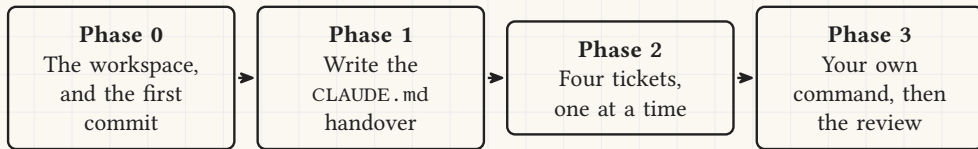
The exercise: Operation Midnight Launch

It is 11 at night. The freelancer who was building the investor demo page has vanished, the investors arrive at nine, and you have one engineer left.



The exercise: Operation Midnight Launch

It is 11 at night. The freelancer who was building the investor demo page has vanished, the investors arrive at nine, and you have one engineer left.



The notebook, and how to open it, are both in the setup guide.

eth-bmai-hs26.github.io/BMAI-PAGE/exercises/we1/operation-midnight-launch.ipynb

Where to read more

- This course's setup guide, with a copy button on every command:
`eth-bmai-hs26.github.io/BMAI-PAGE/guides/claude-code-setup.html`
- The first session, step by step:
`code.claude.com/docs/en/quickstart`

Where to read more

- This course's setup guide, with a copy button on every command:
`eth-bmai-hs26.github.io/BMAI-PAGE/guides/claude-code-setup.html`
- The first session, step by step:
`code.claude.com/docs/en/quickstart`
- Every command:
`code.claude.com/docs/en/commands`

Where to read more

- This course's setup guide, with a copy button on every command:
`eth-bmai-hs26.github.io/BMAI-PAGE/guides/claude-code-setup.html`
- The first session, step by step:
`code.claude.com/docs/en/quickstart`
- Every command:
`code.claude.com/docs/en/commands`
- How it remembers a project:
`code.claude.com/docs/en/memory`

Where to read more

- This course's setup guide, with a copy button on every command:
`eth-bmai-hs26.github.io/BMAI-PAGE/guides/claude-code-setup.html`
- The first session, step by step:
`code.claude.com/docs/en/quickstart`
- Every command:
`code.claude.com/docs/en/commands`
- How it remembers a project:
`code.claude.com/docs/en/memory`



`code.claude.com/docs`